

CS 671: Deep Learning and Applications

Course Code	CS 671
Course Name	Deep Learning and Applications
L-T-P-C	3-1-0-4
Prerequisites	Probability and Random Processes, Linear Algebra
Intended for	B.Tech, MS/PhD in the area.
Distribution	Discipline elective for B.Tech CSE and EE, free elective for others
Approval	15th BoA (Dec 2016); 12th Senate (Mar 2017)

Course Contents

- **Basics of artificial neural networks (ANN):** Artificial neurons, Computational models of neurons, Structure of neural networks, Functional units of ANN for pattern recognition tasks. [4 Lectures]
- **Feedforward neural networks:** Pattern classification using perceptron, Multilayer feedforward neural networks (MLFFNNs), Backpropagation learning, Empirical risk minimization, Regularization, Autoencoders. [6 Lectures]
- **Deep neural networks (DNNs):** Difficulty of training DNNs, Greedy layerwise training, Optimization for training DNNs, Newer optimization methods for neural networks (AdaGrad, RMSProp, Adam), Second order methods for training, Regularization methods (dropout, drop connect, batch normalization). [12 Lectures]
- **Convolution neural networks (CNNs):** Introduction to CNNs convolution, pooling, Deep CNNs, Different deep CNN architectures LeNet, AlexNet, VGG, PlacesNet, Training a CNNs: weights initialization, batch normalization, hyperparameter optimization, Understanding and visualizing CNNs. [12 Lectures]
- **Recurrent neural networks (RNNs):** Sequence modeling using RNNs, Back propagation through time, Long Short Term Memory (LSTM), Bidirectional LSTMs, Bidirectional RNNs, Gated RNN Architecture. [8 Lectures]
- **Generative models:** Restrictive Boltzmann Machines (RBMs), Stacking RBMs, Belief nets, Learning sigmoid belief nets, Deep belief nets. [8 Lectures]
- **Applications:** Applications in vision, speech and natural language processing. [6 Lectures]

Text Books

1. Ian Goodfellow, Yoshua Bengio and Aaron Courville, **Deep learning**, In preparation for MIT Press, Available online: <http://www.deeplearningbook.org>, 2016

References

1. S. Haykin, **Neural Networks and Learning Machines**, Prentice Hall of India, 2010
2. Satish Kumar, **Neural Networks - A Class Room Approach**, 2nd Edition, Tata McGraw-Hill, 2013
3. B. Yegnanarayana, **Artificial Neural Networks**, Prentice- Hall of India, 1999
4. C.M. Bishop, **Pattern Recognition and Machine Learning**, Springer, 2006